

TAMILSELVAN G

Software Engineer | .NET Full Stack Developer | Angular | ASP.NET Core

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of experience building enterprise web applications with Angular (14–20), ASP.NET Core (.NET 6, .NET 8 and .NET 10) and SQL Server. Develops REST APIs and Angular frontend and sees modules through from first setup to deployment and production support. Has worked on publishing, workforce management, project management and email-monitoring systems, including dashboards that display AI-generated email summaries and classifications.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, Python

.NET & Backend: ASP.NET Core Web API, ASP.NET MVC with Razor Views (.NET Framework 4), .NET 6, .NET 8, .NET 10, Entity Framework Core, LINQ, Dependency Injection, Repository Pattern, Unit of Work, SOLID Principles, Clean Architecture, Middleware, REST APIs, API Integration, SignalR

Security: Authentication & Authorization, JWT Authentication, Role-Based Access Control (RBAC)

Frontend: Angular 14–20, Components, Services, Directives, Pipes, Routing, Route Guards, HTTP Interceptors, HTTP Client, RxJS Observables, Signals, State Management, Reactive Forms, Lazy Loading, HTML5, CSS3, Tailwind CSS, Bootstrap, Angular Material, Apex Charts, TinyMCE

Databases: SQL Server (Stored Procedures, SQL Query Optimization, Indexing), MongoDB, Redis

Cloud, DevOps & Hosting: Docker, Docker Compose, Git, GitHub Actions, AWS S3, IIS

API Tools & Debugging: Postman, Swagger/OpenAPI, Chrome Dev Tools, API Testing, Troubleshooting

AI & Integration: RAG, LLM Integration, Google Gemini API, Ollama, Vector Embeddings, Qdrant, SSE Streaming, Prompt Engineering

Practices: Agile/Scrum, Code Review, Performance Optimization

PROFESSIONAL EXPERIENCE

Nova Techset Pvt. Ltd. - Chennai, India

Associate Software Engineer (Sep 2024 – Present) | Junior Software Engineer (Mar 2023 – Aug 2024) | Trainee

Software Engineer (Jun 2022 – Feb 2023)

- Built and maintained full-stack features across five enterprise applications covering publishing workflows, content editing, workforce management, project management and employee profiling, and grew from Trainee to Associate Software Engineer along the way.
- Developed REST APIs in ASP.NET Core Web API on .NET 6, .NET 8 and .NET 10 using Entity Framework Core, LINQ and Dependency Injection. Followed SOLID principles and used the Repository Pattern with Unit of Work for data access.
- Maintained and extended an ASP.NET MVC application on .NET Framework 4, working with Razor views, controllers and models.
- Wrote SQL Server stored procedures and tuned slow queries through indexing and SQL query optimization. Together with Redis caching, this improved backend performance by roughly 15% at project level.
- Implemented authentication and authorization with JWT and role-based access control (RBAC) so each user role sees only its own screens and data.
- Built Angular 14–20 front ends from reusable components, services, custom directives and pipes, with routing protected by route guards, HTTP interceptors on API calls, Reactive Forms, and Signals and RxJS Observables for state management.
- Improved Angular performance with lazy loading, RxJS optimization and frontend profiling.
- Added real-time collaboration features with SignalR for the publishing and workflow applications.
- Used MongoDB for the project management and email-processing workflows, including conversation data, message tracking and dashboard queries.
- Containerized applications with Docker and Docker Compose and supported hosting and deployment on IIS.
- Tested and debugged APIs with Postman, Swagger/OpenAPI and Chrome DevTools, and worked with the AI team on API integration of AI-generated email summaries and classification results into enterprise dashboards.

KEY PROJECTS

1. nPMS - Project Management & Email Monitoring System

Technologies: Angular 20, TypeScript, ASP.NET Core (.NET 10), C#, MongoDB, REST API, JWT, AI Integration

- Own the front end and back end of an enterprise system that tracks project-related emails sent to project managers.
- Built the classification workflow that sorts incoming emails by business rules, with role-based dashboards for management, publishers, managers, team leaders and project managers.
- Added tracking for pending responses and for responses made after 20 hours, so SLA slippage is easy to spot.

- Integrated AI-generated email summaries and classification reasons from the AI team's processed data into the application.
- Designed the MongoDB structure for email conversations, covering received and sent messages and conversation-level status.

2. nVMS - Vendor Management System

Technologies: Angular, TypeScript, Tailwind CSS, *ASP.NET Core (.NET 6)*, C#, SQL Server, JWT, REST API

- Developed and maintained a vendor management platform for publishing workforce operations, managing a network of 300+ freelancers.
- Built the front-end modules for vendor onboarding, qualification, service allocation and invoice management, plus vendor competency and assessment workflows. reducing administrative overhead by **30%**
- Worked on invoice and billing workflows that depend on word counts, deliverable types and rates.
- Secured the application with JWT authentication and role-based authorization and used caching with optimized database operations for frequently called APIs.
- Integrated the team's existing AWS S3 file upload and download module into the application.

3. nEDIT - Web-Based Content Editing & XML Proofing Platform

Technologies: Angular, TypeScript, TinyMCE, ICE Plugin, SignalR, *ASP.NET Core (.NET 6)*, C#, SQL Server, XML

- Contributed to an enterprise application used for content editing, XML validation and proofing in publishing operations.
- Developed rich-text editing with TinyMCE, including formatting, paragraph styles, footnotes and image operations.
- Implemented and enhanced change tracking with the ICE Plugin for revision history and content comparison.
- Worked on SignalR features that keep editing and workflow updates in sync between users.
- Built support for mathematical content, spell-checking and XML validation while keeping XML structure intact through editing and proofing.

4. Sesame - Article Tracking & Publishing Workflow System

Technologies: Angular, TypeScript, ASP.NET Web API 2, .NET Framework 4, C#, SQL Server, REST APIs

- Worked on an enterprise publishing application used to track articles and manage their workflow through different stages.
- Developed and maintained multiple master pages used to configure and manage application-level business data.
- Worked across frontend and backend layers to implement business requirements and connect Angular screens with ASP.NET Web API 2 endpoints. Handled UI development, API integration, form validation and data management for assigned modules.

PERSONAL PROJECT

AI RAG Assistant - Document Question-Answering Application

Technologies: Python, FastAPI, Qdrant, Redis, Google Gemini API, Ollama, Docker, GitHub Actions, Render

- Built a personal Retrieval-Augmented Generation (RAG) application that allows users to upload PDF, DOCX, TXT and CSV documents and ask questions against their content.
- Implemented document ingestion, text extraction, chunking, embedding generation and vector search using Python, Qdrant and Google Gemini.
- Used Qdrant for vector storage and similarity search to retrieve relevant document chunks before generating answers.
- Integrated Google Gemini for LLM-based responses and used Ollama with local models such as bge-m3 and qwen2.5:7b during local development and experimentation.
- Used Redis for conversation memory and application-level caching.
- Built REST APIs with FastAPI and organized the application using dependency-based services to keep the components easier to maintain and test.
- Containerized the application using Docker Compose and created a GitHub Actions CI/CD workflow for deployment to Render.

EDUCATION

Bachelor of Engineering - Computer Science and Engineering

Alpha Engineering College, Anna University | 2021 | CGPA: 7.45 / 10